Machine Learning

probabilistic view point.

X : domain set (Input)

Y : Range set (output)

Given pairs $D = \{(x_1, y_1), (x_2, y_2), \ldots (x_N, y_N)\}$

where x_i, y_i is a pair of observation.

Assume that $\exists$ a function $f : X \to Y$

$f(x_i) = y_i$

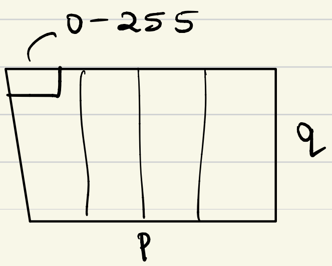
The underlying function "f" is unknown.

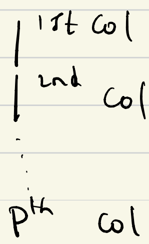
$\boxed{\text{Given } D, \text{ find } f}$

start with some initial guess on f.

"refine" the guess using D (data)

Consider X_i to be x-ray images.





Pq-sized vector.

$\mathbb{R}^{pq}$ – a point in $\mathbb{R}^{pq}$ dim.

$$x_i \in X \subset \mathbb{R}^d$$

$$y_i \in \{0, 1\}$$, $0 \rightarrow$ diseased
$1 \rightarrow$ benign / non-diseased.

$$f : X \rightarrow Y$$

"f" is complex to be estimated from the physics of the problem.

Resort to statistical methods:
make repeated observations & estimate f.

In a probabilistic frame work

X , Y define Random Variables.

Ω : sample space [set of all possible outcomes of a random Experiment]

Consider the subsets of sample space.
Let $\mathbb{F}$ denote the collection of all possible subsets.
Objective: Assign a "measure" on $\mathbb{F}$.

Probability measure is one such measure.

$$\mathbb{P} : \mathbb{F} \longrightarrow [0, 1]$$

properties of $\mathbb{P}$.

if $A, B \in \mathbb{F}$.

i) $\mathbb{P}(A) \geq 0 \quad \forall A \in \mathbb{F}$.

ii) $\mathbb{P}(\Omega) = 1$, $\mathbb{P}(\phi) = 0$

iii) A, B s.t $A \cap B = \phi$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$

$(\Omega, \mathbb{F}, \mathbb{P})$: Prob. Triplet.

Define a function X (random variable) from Ω to $\mathbb{R}$.

$$X : \Omega \longrightarrow \mathbb{R}.$$

$\Omega \rightarrow \mathbb{R}$

$\mathbb{F} \rightarrow$ B-sigma algebra. , $(-\infty, x]$

$\mathbb{P} \rightarrow \mathbb{P}_x$: distribution function.

$$\mathbb{P}_x(x) \triangleq \mathbb{P}\left[A : X^{-1}(-\infty, x] \right]$$

$(\Omega, \mathbb{F}, \mathbb{P}) \xrightarrow{\text{R.v. } X} \underline{(\mathbb{R}, \mathcal{B}, \mathbb{P}_x)}$: work with this.

Random varibles with vector-valued range spaces.

In general, R.V. have $\mathbb{R}^d$ as their range spaces.

$$X : \Omega \longrightarrow \mathbb{R}^d \quad \text{(vector-valued R.V.)}.$$

$\mathbb{P}_{\underline{X}}(X \in \mathbb{R}^d)$ = probability of the inv. image of cartisian product under X.

Vector valued RVs.

Here the range set of the function (RV), is $\mathbb{R}^d$ where d is a scalar.

$$X : \Omega \longrightarrow \mathbb{R}^d.$$

Joint distributions :

Let Ω be a sample space.

Define two functions, x_1 & x_2

$$x_1 : \Omega \longrightarrow \mathbb{R} \, , \quad \mathbb{P}_{x_1}$$
$$x_2 : \Omega \longrightarrow \mathbb{R} \, , \quad \mathbb{P}_{x_2}$$

Define Joint probability distribution as

$$\mathbb{P}_{x_1,x_2}(a,b) = \mathbb{P}\left[E : \text{intersection of inverse images of } (-\infty,a] \text{ \& } (-\infty,b] \text{ under } x_1 \text{ \& } x_2, \text{ respectively}\right]$$

The above idea can be extended to d scalar Random variables.

Define Conditional probabilities / distributions.

If A & $B \subset \mathbb{F}$,

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Suppose X & Y are two RVs defined on Ω,

$$\mathbb{P}_{X|Y}(x|Y=y) = \frac{\mathbb{P}_{XY}}{\mathbb{P}_Y}$$

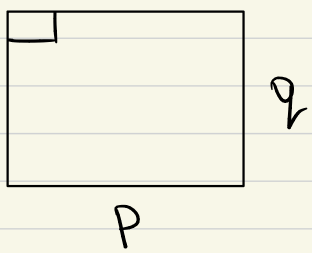
Marginal distribution : If X & Y are two RVs, marginal of X is defined as

$$\mathbb{P}_X = \int_{\text{range}(Y)} \mathbb{P}_{XY} \, dy$$

$$\mathbb{P}_Y = \int_{\text{range}(X)} \mathbb{P}_{XY} \, dx$$

Example:

Consider an image of dimensions $p \times q$.



can be viewed as a $\mathbb{R}^{pq}$-dim vector.

In general, any datapoint is a d-dim vector

Every datapoint is an element in the range space of a random variable. X

The distribution function indicates / quatifies the likelihood of observing a datapoint under X.

Distribution function completely specifies the sample space.

Typically, a "label" is taken to be another additional random variable defined on the same sample space.

In this scenario, we have two RVs, X & Y, $X \in \mathbb{R}^d$ & $Y \in \mathbb{R}^k$

X : dataspace Y : label space $\{1, 2, 3, 4, 5\}$

The convenction is as follows:

"Learning" starts with a dataset

$$D = \{ (x_1, y_1), (x_2, y_2), (x_3, y_3) \ldots (x_N, y_N) \}$$

$\sim$ i.i.d $\mathbb{P}_{XY}$

Independent

Identically distributed

Independent accross data points & not data-dimensions.

All problems in Machine learning can be seen as

$\boxed{\text{Given } D \sim \text{unknown } \mathbb{P}_X, \text{ Estimate } \mathbb{P}_X \text{ or conditionals on } \mathbb{P}_X \text{ \& sample from it}}$

Recall,

Given a dataset

$$D = \{(x_i, y_i)\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_{XY} \text{ (unknown)}$$

$$x_i \in \mathbb{R}^d, \quad y_i \in \mathbb{R}^K \text{ or } \{1, 2, \ldots, K\}$$

Typically, X : input / features / data
Y : label / output.

X & Y are random variables defined on a sample space.

Fundemental problem of ML :

Given $D \sim$ unknown distribution,

a) Estimate the distribution
b) Sample from the distribution

Distribution Estimation

Given $D = \{(x_i, y_i)\} \sim \text{iid } \mathbb{P}_{xy}$

Examples :

i) Estimate $\mathbb{P}_{y|x}$: Classification , $y \in \{1, 2..K\}$

$p_{y|x}$ Regression , $y \in \mathbb{R}^K$

ii) Estimate $\mathbb{P}_x$, $\mathbb{P}_y$ or $\mathbb{P}_{xy}$, $\mathbb{P}_{x|y}$

p_x

In all conditionals, $\mathbb{P}_{x|y}\left(x \middle| y = y\right)$

Probability density function:

Give a CRV with a dist. function $\mathbb{P}_x$,
density function $p_x : X \to \mathbb{R}^+$, s.t

$$\mathbb{P}_x(x) = \int_{-\infty}^{x} p_x(x)\, dx$$

The Challenge with ML

Given $D \sim$ iid from unknown p_x , Estimate p_x.

Challenge : $\mathbb{P}_x$ is completely unknown.

<u>Question</u> : How to estimate a density function given its samples?

Consider we have a dataset D.

$$D = \{x_i\}_{i=1}^{N} \sim \text{iid } p_x$$

i) Assume a parametric functional form on p_x, denoted by p_θ. "Model Choice"

$$x \in \mathbb{R}^d$$

Eg: $p_\theta^a(x) = W_1^T x + W_2$, $\quad W_1 \,\&\, W_2 \in \mathbb{R}^d$

$$\theta = \{W_1, W_2\}$$

$$p_\theta^b(x) = \mathcal{N}(x; \mu, \Sigma)$$

$$\theta = \{\mu, \Sigma\}, \quad \mu \in \mathbb{R}^d, \; \Sigma \in \mathbb{R}^{d \times d}$$

θ: parameters.

ii) Define or <u>compute</u> a distance metric between p_x & p_θ.

Let d denote the "distance metric" b/w p_x & p_θ.

$$d(p_x, p_\theta) : P_x \times P_\theta \rightarrow \mathbb{R}^+$$

iii) Find | Estimate the parameters θ by solving the following optimization problem.

$$\boxed{\theta^* = \arg\min_{\theta} d\left(p_x, p_\theta\right)}$$ "Training the model".